

DANA 4800

Homework #1

Due: 11:59 pm on September 21st , 2025
Submitted to: BrightSpace Homework Folder

Please read these instructions before submitting your homework.

1. **Format:** Please convert all work to PDF.
2. **R Output:** If you have R Output, please copy and paste the output to the corresponding part of the Word document.
3. **Submission:** No email submission will be accepted regardless of the circumstances.
4. **Number of Submission:** You are allowed to submit your homework as many times as you want. But only the latest version of your submission will be marked.
5. **Order of Work:** All work must be submitted in the same order of the questions. It is suggested that you use a brand new page for a different question, especially true for hand-written work.
6. **Filename:** Please name your file in the following format:
"DANA4800_Lastname_Firstname_HW#.pdf". For example, my first assignment would have the name "DANA4800_Lo_Michael_HW1.pdf".
7. **Last Advice:** Internet connections do go off, computers do break down, or other unexpected events do occur when they are least expected. The onus is squarely on you to submit it on time. No excuses.

Note: Failure to comply with any of the above will result in loss of marks.

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1. An HR manager at XY Corporation wanted to find out the number of overtime hours employees typically work in 2025. A random sample of 60 employees was drawn in first quarter of 2025 to investigate this.
 - a) Identify the researcher. **[1 mark]**
 - b) Provide a description of the objective. **[1 mark]**
 - c) Identify the subjects of interest. **[2 marks]**
Note: Make sure you also include the when and where, if available.
 - d) Provide a description of the variable of interest. Please also provide the units of measurement if it is a numerical variable or list out all possible classes if it is a categorical variable. **[1+1 marks]**
 - e) Identify the type of the variable and the corresponding scale of measurement. **[1+1 marks]**
Note: Marks will be deducted with missing words (like “variable” or “scale”).
 - f) Provide a description of the population of interest. **[1 mark]**
 - g) Identify the sampling method used by the researcher. **[1 mark]**
 - h) Provide a description of the sample. **[1 mark]**
 - i) Is there an issue with the selection bias? Briefly justify your answer by comparing the (target) population and the sampling frame. **[0+2 marks]**
2. Sophia lives in Calgary. She was thinking about buying a new smartphone and wanted to know what percentage of people in her city own an iPhone. She asked 30 of her friends and relatives if they used an iPhone or not to make her decision.
 - a) Identify the researcher. **[1 mark]**
 - b) Provide a description of the objective. **[1 mark]**
 - c) Identify the subjects of interest. **[2 marks]**
Note: Make sure you also include the when and where, if available.
 - d) Provide a description of the variable of interest. Please also provide the units of measurement if it is a numerical variable or list out all possible classes if it is a categorical variable. **[1+1 marks]**
 - e) Identify the type of the variable and the corresponding scale of measurement. **[1+1 marks]**
Note: Marks will be deducted with missing words (like “variable” or “scale”).
 - f) Provide a description of the population of interest. **[1 mark]**
 - g) Identify the sampling method used by the researcher. **[1 mark]**
 - h) Provide a description of the sample. **[1 mark]**
 - i) Is there an issue with the selection bias? Briefly justify your answer by comparing the (target) population and the sampling frame. **[0+2 marks]**
3. The Chairperson of Vancouver schooling board wanted to know the average number of days high school students were absent during the first week of the fall semester, 2024, in Vancouver, BC. He divided the whole city into 4 parts- North, South, East. He picked one school from each part and took data of 20 random students from each school to get his sample.
 - a) Identify the researcher. **[1 mark]**
 - b) Provide a description of the objective. **[1 mark]**
 - c) Identify the subjects of interest. **[2 marks]**
Note: Make sure you also include the when and where, if available.
 - d) Provide a description of the variable of interest. Please also provide the units of measurement if it is a numerical variable or list out all possible classes if it is a categorical variable. **[1+1 marks]**

- e) Identify the type of the variable and the corresponding scale of measurement. **[1+1 marks]**

Note: Marks will be deducted with missing words (like “variable” or “scale”).

Note: I suggest you provide some reasoning to justify your choice of the type of variable.

- f) Provide a description of the population of interest. **[1 mark]**
- g) Identify the sampling method used by the researcher. **[1 mark]**

- 4. Students are generally confused with the stratified random sampling method and the cluster sampling method. I hope students have a better understanding of them after completing this question.

There are two situations in the questions: one method is used in each of the two situations. In each of the following two situations, (1) identify the sampling method used, (2) describe how you would define the non-overlapping groups, and (3) outline the three main steps.

An Economist wanted to find out the average number of hospital visits, because of an injury, people had in 2024 in the city of Vancouver.

- a) He thought of dividing the population into 8 age groups- 15 years and below, 15-25, 25-35, 35-45, 45-55, 55-65, 65-75, and 80 years and above. He would then call 30 random people from each group to make a sample of 240. **[1+1+3 marks]**
- b) He also thought of dividing the city of Vancouver into 20 smaller non- overlapping geographical sub-parts. He would then choose 5 random ‘sub-parts’ and take all of the subjects in them to make his sample. **[1+1+3 marks]**

- 5. A college has 40 classrooms, each with 25 students. The student union is organizing a survey about campus food services. Identify the most appropriate sampling method used in each case below.

- a) The first 100 students who arrive at the cafeteria are asked to complete the survey. **[1 mark]**
- b) From each classroom, three students are randomly chosen to participate in the survey. **[1 mark]**
- c) A complete list of all students enrolled at the university is prepared. From this list, 100 students are randomly chosen for the survey. **[1 mark]**
- d) Five classrooms are randomly selected, and all students in those classrooms are surveyed. **[1 mark]**
- e) The union puts up an online survey for any student to fill. And the students who participate make up the sample. **[1 mark]**
- f) A student is randomly chosen from the first 10 who arrive at the cafeteria, and then every 10th student thereafter is surveyed. **[1 mark]**

- 6. General public underestimate the power of DANAistics. In particular, misuse of DANAistics could have a devaDANAing impact on individuals or organizations. Your task is to do a Google search about the 1936 US Presidential Election between the incumbent Democratic candidate Franklin D. Roosevelt and Republican candidate Alf Landon. You want to pay special attention to the pre-election prediction between the two organizations – Literary Digest and Gallup Poll. Answer the following questions.

Note: You might get slightly different values from different websites. So, there are no standard answers here.

- a) What was the success rate (predicting the correct election outcome) of *Literary Digest* prior to 1936? **[1 mark]**

b) What were the 1936 pre-election prediction by both *Literary Digest* and *Gallup Poll*? Please focus on your answer as 1) who would win the 1936 Presidential election and by what percentage of popularity vote. **[2 marks]**

c) What was the official result of the 1936 US Presidential Election? **[1 mark]**

Now let us focus on the *Literary Digest* only from this point on.

d) In relation to selection bias, what did the *Literary Digest* do (or not do) to wrongly predict the election results? Please provide as much details as possible. **[2 marks]**

e) In relation to non-response bias, what did the *Literary Digest* do (or not do) to wrongly predict the election results? Please provide as much details as possible. **[2 marks]**

f) In relation to response bias, what did the *Literary Digest* do (or not do) to wrongly predict the election results? Note that The Great Depression started in 1929 and last till around 1939. So, general public did not know when the Great Depression would end. So, imagine to whom you would point your fingers when time is tough. **[2 marks]**

g) Now focus on modern days. What are your thoughts about 2016 US Presidential Election between Democratic candidate Hilary Clinton and Republican candidate Donald Trump, in relation to the three biases that we have learned? **[2 marks]**

This is the end of HW1